Asymmetric Information

one party knows more than another in a transaction

two types of problems:

1) moral hazard: one party can take hidden actions that affect risk (the value of an agreement)
(after an agreement is made)

ex. individual gets full insurance on his car (ex. from last class)

now drives more agressively, take less precaution against theft...

if insurance was priced "fairly" initially, now risk ↑ → insurance company loses money on avg.

solutions: increase premiums if accident

less than "full insurance" (deductible: you pay this if an accident happens)

* these move some risk back to the insured party

2) adverse selection: one party has hidden type that affects risk
(before contract is signed)

The market for Lemons (Akerlof, 1970)

Asymmetric information can cause market collapse

suppose there are 100 people wanting to sell their cars, and 100 people wanting to buy

	sellers WTA	buyers WTP
25 cars are top quality ("peach")	4000	5000
25 cars are good quality	3000	3500
25 cars are fair quality	2000	2500
25 cars are poor quality ("lemon")	1000	1200

buyers' willingness to pay (WTP) is higher than sellers' willingness to accept (WTA) for each quality type

→ all cars would be sold and market "clears"

key: sellers know the quality but buyers do not

→ with asymmetric information, buyers need to "guess" quality

suppose buyers are willing to pay expected value

$$EV = \left(\tfrac{1}{4}\right)(5000) + \left(\tfrac{1}{4}\right)(3500) + \left(\tfrac{1}{4}\right)(2500) + \left(\tfrac{1}{4}\right)(1200) = 3050$$

but, who is willing to sell at that price?

lemons, fair and good → sellers of "peaches" won't → leave the market

buyers realize this and must accept/update their expectations

now, $EV = \left(\tfrac{1}{3}\right)(3500) + \left(\tfrac{1}{3}\right)(2500) + \left(\tfrac{1}{3}\right)(1200) = 2400$

again, at this price, good quality sellers exit the market

now, $EV = \left(\tfrac{1}{2}\right)(2500) + \left(\tfrac{1}{2}\right)(1200) = 1850$

→ fair quality sellers leave the market at this price

equilibrium price: between 1000 and 1200 ⇒ only lemons are sold

<u>ex.</u> Health insurance: people differ in their risk of requiring expensive medical treatment

if insurance company offers policy that is priced based on average risk in population

low risk people don't buy

prices have to adjust to reflect true risk

moderate risk people don't buy

Solutions?

"menu" of prices with different deductibles ("co-pays")

mandatory insurance (universal healthcare)

health screening/denying coverage for "pre-existing conditions"

Final Exam

cumulative → weighted a little bit towards later material (after midterm)

5 questions

1. short explanation
2. short calculation
3. longer calculations
4. longer calculations
5. longer calculations